SECTIONS WITH COMPACT SUPPORT IN DEGREE ZERO ARE THE GLOBAL SECTIONS VANISHING NEAR THE COMPLEMENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0G5C (sources/stacks-duality/pinned-original-source.tex line 9276).

Allowed standard prerequisites (editorial).

- The covariance map for the lower shriek and compactly supported cohomology of an open. Same chapter, Remark labelled remark-covariance-open-lower-shriek, cited at proof lines 9294-9296. Provides the map whose image is described.
- Artin-Rees for the powers of the ideal sheaf. Cohomology of Schemes, Lemma labelled coherent-lemma-Artin-Rees, cited at proof lines 9302-9304. Identifies the limit of the complexes with the limit of the H^0(X, I^n F).
- Krull intersection for a finite module over a Noetherian ring. Algebra, Lemma labelled algebra-lemma-intersect-powers-ideal-module-zero and Remark labelled algebra-remark-intersection-powers-ideal, cited at proof lines 9308-9312. Characterises the sections contained in every I^n F as those vanishing near Z.

Source reference. Original reference 0G56: sources/stacks-duality/pinned-original-source.tex, source lines 8923–8923. Same-chapter pointer into the author environment remark at source line 8923; invoked at proof line 9292 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 01YA: sources/stacks-chow/tags.txt, source lines 2549–2549. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9301 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00IP: sources/stacks-chow/tags.txt, source lines 687–687. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9308 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 00IR: sources/stacks-chow/tags.txt, source lines 689–689. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9310 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let X be a proper scheme over a field k. Let $K \in D^b_{Coh}(\mathcal{O}_X)$ with $H^i(K) = 0$ for $i < 0$. Set $\mathcal{F} = H^0(K)$. Let $Z \subset X$ be closed with complement $U = X \setminus U$. Then*

$$H^0_c(U, K|_U) \subset H^0(X, \mathcal{F})$$

is given by those global sections of $\mathcal{F}$ which vanish in an open neighbourhood of Z.

Proof. Consider the map $H^0_c(U, K|_U) \to H^0_X(X, K) = H^0(X, K) = H^0(X, \mathcal{F})$ of Remark 0G56. To study this we represent K by a bounded complex $\mathcal{F}^\bullet$ with $\mathcal{F}^i = 0$ for $i < 0$. Then we have by definition

$$H^0_c(U, K|_U) = \lim H^0(X, \mathcal{I}^n\mathcal{F}^\bullet) = \lim \mathrm{Ker}(H^0(X, \mathcal{I}^n\mathcal{F}^0) \to H^0(X, \mathcal{I}^n\mathcal{F}^1))$$

By Artin-Rees (Cohomology of Schemes, Lemma 01YA) this is the same as $\lim H^0(X, \mathcal{I}^n\mathcal{F})$. Thus the arrow $H^0_c(U, K|_U) \to H^0(X, \mathcal{F})$ is injective and the image consists of those global sections of $\mathcal{F}$ which are contained in the subsheaf $\mathcal{I}^n\mathcal{F}$ for any n. The characterization of these as the sections which vanish in a neighbourhood of Z comes from Krull's intersection theorem (Algebra, Lemma 00IP) by looking at stalks of $\mathcal{F}$. See discussion in Algebra, Remark 00IR for the case of functions. □